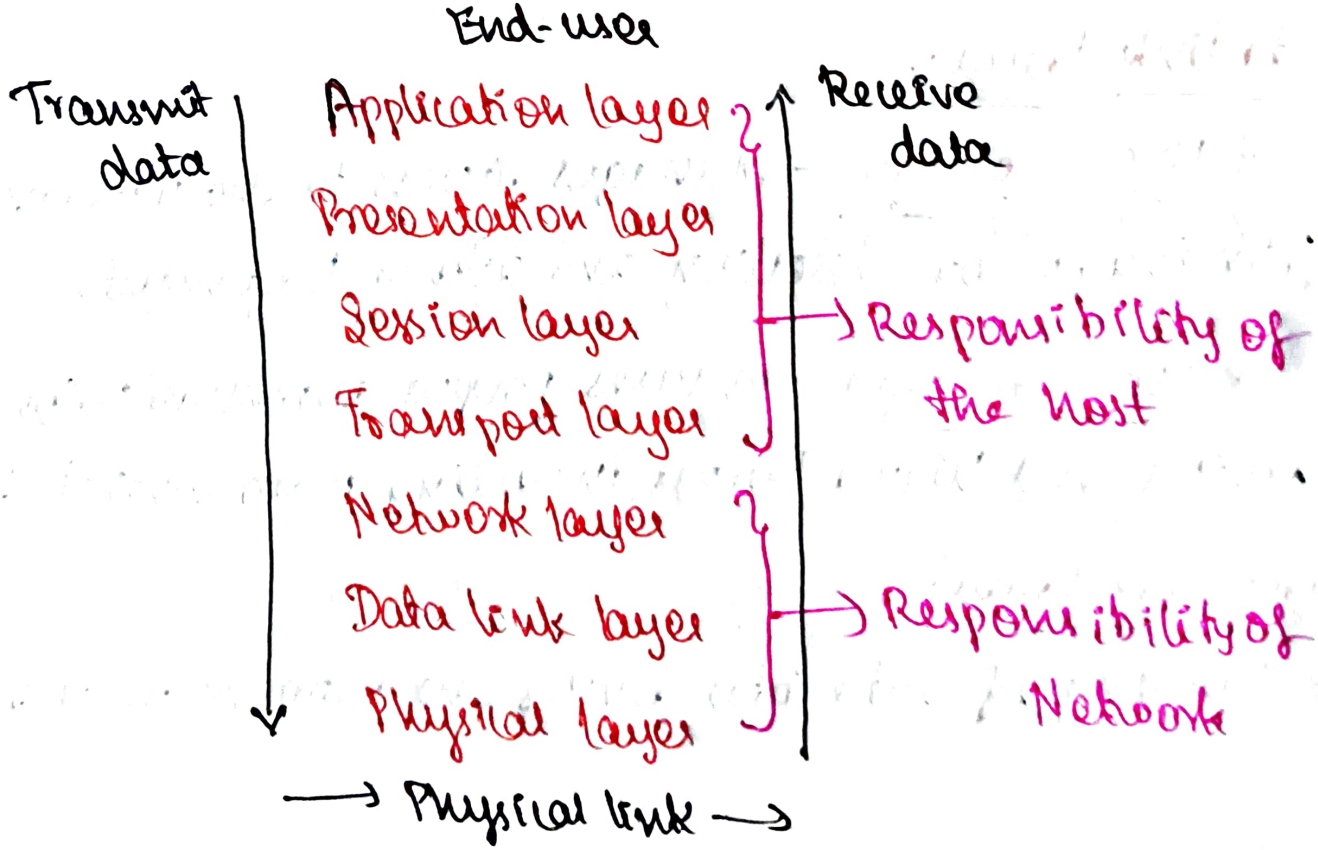
2) OSI NETWORK ARCHITECTURE / OSI MODEL:-

* The OSI (open system interconnection) model is a framework that standardizes how different network protocols interact in seven layers.

* Each layer has specific responsibilities in ensuring communication across a network.



1) Physical layer:

* It deals with the transmission of raw data over physical media like cables or wireless signals.
* It converts data into signals such as electrical, light or radio signals.
* It defines how devices on a network are physically connected and how to transmit data.

Eg: voltage levels, data rates.

II) Data link layer:

* It ensures that data is transferred reliably between two devices on the same network.
* Framing: Breaks data from the upper layers into smaller units called frames or transmission.
* Error detection & correction: Detects errors that might occur during transmission and request retransmission if necessary.
* Flow control: controls the pace of data transfer to prevent congestion.

III) Network layer:

* It determines how data is routed across different networks and handles addressing (IP).
* Routing: It determines the best path for data to travel from one device to another, potentially across multiple networks.

* Logical addressing: uses IP addresses to identify devices on a larger networks, such as the internet.

* Packet forwarding: data is divided into smaller units called packets and sent to the destination based on the routing process.

---

iv) Transport layer:

* It ensures reliable data transfer between devices, manages error correction and control flow.

* Segmentation & reassembly: breaks large chunks of data into smaller segments for easier transmission & reassembles them at the receiving end.

* Reliability: ensures reliable delivery of data through error detection, error correction and retransmission of lost packets (if needed).

---

v) Session layer:

* manages sessions (connections) between applications, ensuring they stay open and handle communication properly.

* dialog control: manages whether the communication is half duplex (one way at a time) or full duplex (both ways at the same time).

* Synchronization: ensures that data is sent and received in the correct order & can synchronize data if needed.

Eg: resuming file transfers.

vi) Presentation layer:

* Translates data between the application layer and the network, ensuring data is in a usable format.
* data translation: translates data between different formats.

  Eg:- from ASCII to EBCDIC

* Data encryption: Encrypts data for security during transmission and decrypts it at the receiving end.
* Data compression: compresses data to reduce the size of transmission and decompresses it for the recipient.
* Data encoding - encodes and decodes data

  Eg:- converting images, text etc. to a suitable format.

vii) Application layer:

* Provides network services directly to end users or applications like email, file transfer, web browsing.
* Application interfaces: provides the interface between the end-user and the network.
* Network services: delivers services like file transfer (FTP), web browsing (HTTP), email (SMTP) and more.
* user interaction: provides the environment for users to interact with applications.

  Eg:- browsers, email clients.